MCGI: Manifold-Consistent Graph Indexing for Billion-Scale Disk-Resident Vector Search

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Abstract

Graph-based Approximate Nearest Neighbor (ANN) search often suffers from performance degradation in high-dimensional spaces due to the Euclidean-Geodesic mismatch, where greedy routing diverges from the underlying data manifold. To address this challenge, this paper presents Manifold-Consistent Graph Indexing (MCGI), a geometry-aware and disk-resident indexing method that leverages Local Intrinsic Dimensionality (LID) to dynamically adapt search strategies to the intrinsic geometry of data. Unlike conventional algorithms that treat dimensions uniformly, MCGI modulates its beam search budget based on in-situ geometric analysis, which reduces sensitivity to data-specific hyperparameters by replacing a single scalar with a geometry-informed range that remains stable across datasets of varying dimensionality. Theoretical analysis demonstrates that MCGI provides robust approximation by preserving manifold-consistent topological connectivity. At matched high recall, MCGI raises the query throughput of DiskANN on the same index by up to $1.43\times$ on three million-scale benchmarks and by up to $1.24\times$ on the billion-scale SIFT1B and Deep1B.

1 Introduction

The advent of Large Language Models [5, 36] has fundamentally transformed the landscape of information retrieval and knowledge management. To address the inherent limitations of LLMs, such as hallucinations [20] and knowledge cutoff dates, Retrieval-Augmented Generation (RAG) [26] has emerged as a critical architectural paradigm. RAG relies heavily on the ability to retrieve semantically relevant context from massive corpora in real-time, typically via dense vector representations [24]. This dependency has placed Approximate Nearest Neighbor Search (ANNS) at the core of modern data infrastructure, demanding vector indices that can scale to billion-point datasets [32] while maintaining low latency and high recall under strict production constraints.

For extreme-scale embedding vector applications, many ANNS techniques have noticeably converged on graph-based indices, with DiskANN (Vamana) and its variants [33, 6, 12, 13] being representative solutions for large-scale, SSD-resident workloads. These algorithms typically employ greedy routing on a proximity graph to navigate from an entry point to the query target. While such methods exhibit exceptional performance on popular benchmarks like SIFT1M [21] (128 dimensions), their efficiency degrades significantly in high-dimensional spaces, such as GIST1M (960 dimensions). This degradation is often attributed to the curse of dimensionality [3], where the distance contrast diminishes, and the Euclidean shortest path on the graph diverges from the geodesic path on the underlying data manifold [34, 19]. We refer to this phenomenon as the *Euclidean-Geodesic Mismatch*, or EGM. When the routing algorithm ignores the intrinsic geometry of the data, it performs excessive backtracking and disk I/O, rendering the search inefficient, which is particularly problematic in high-recall scenarios where production systems often expect Recall@10 higher than 95%.

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Our observation on this EGM problem is that high-dimensional real-world data is rarely uniformly distributed; Instead, it typically adheres to the Manifold Hypothesis [34, 31], residing on lower-dimensional structures embedded within the ambient space. Consequently, the search difficulty is not uniform across the dataset but is modulated by the Local Intrinsic Dimensionality (LID) [25]. That is, in regions where the data manifold is flat (low LID), greedy routing is effective; however, in regions with high curvature or complex topology (high LID), standard greedy strategies may fail to identify the descent direction. *We thus argue that an optimal indexing strategy must be manifold-aware, dynamically allocating computational resources based on the local geometric complexity.*

To demonstrate the above hypothesis, we design Manifold-Consistent Graph Indexing (MCGI), a geometry-aware disk-based indexing method crafted to align Euclidean search with the underlying data manifold. By incorporating LID estimation directly into the routing logic, MCGI demonstrates the ability to adapt its traversal strategy to the local topology of the underlying manifold.

The remainder of this paper will present the following contributions: (1) A theoretical framework embedding the local intrinsic dimensionality to graph navigability with formal guarantees on estimation robustness, fixed-beam suboptimality, and construction-time adaptation; (2) Geometry-aware algorithms that adapt the pruning parameter of each node during construction and the candidate list of each query during search; and (3) up to $1.33\times$, $1.43\times$, and $1.20\times$ higher throughput than DiskANN at matched high recall on SIFT1M, GloVe-100, and GIST1M, respectively, and up to $1.22\times$ and $1.24\times$ on one billion vectors of SIFT1B and Deep1B, respectively.

2 Related Work

Vector Indexing Paradigms While traditional sparse retrieval methods rely on lexical matching [30], memory-resident graph indices like HNSW [28] establish state-of-the-art dense retrieval but face billion-scale memory bottlenecks. Disk-based approaches, such as DiskANN [33], SPANN [6], PipeANN [12], and OdinANN [13], along with predecessors like NSG [10], adapt topologies or inverted structures for SSDs. Recent advancements such as CSPG [38] further optimize sparse proximity graphs to reduce unpromising explorations, yet these methods predominantly rely on static routing parameters evaluated on moderate dimensionality. In contrast, MCGI dynamically modulates the search with local topology to align graph traversal with the underlying manifold.

High-Dimensional Indexing Early space-partitioning trees like the KD-tree [4] and R-tree [15] severely degrade to linear-scan performance in dimensions higher than 20 due to the curse of dimensionality. Approximate methods address this through sub-linear hashing like LSH [18, 8], which incurs excessive storage overhead, or subspace quantization like PQ [21] and OPQ [11], and randomized structures like RP-Trees [7], which sacrifice accuracy through compression. Furthermore, recent literature extensively studies the theoretical limits of navigable graphs [9] and the efficiency of index construction [40] in strictly high-dimensional scenarios. MCGI circumvents aggressive quantization losses and redundant hashing overheads by preserving the exact connectivity benefits of graph traversal adapted to local topology.

Intrinsic Dimensionality Theoretical metrics such as the doubling dimension [14] and expansion dimension [23] provide asymptotic bounds on search complexity, and recent analyses establish worst-case performance limits for popular implementations like DiskANN and HNSW based on intrinsic dimensionality [19]. Maximum likelihood estimators [25] for Local Intrinsic Dimensionality (LID) enable robust practical estimations. Existing works typically utilize LID passively for query hardness prediction [16, 37] or detecting adversarial examples [27] without altering the underlying index structure. Diverging from passive descriptive applications, MCGI actively utilizes LID as a prescriptive control signal to dynamically modulate graph traversal parameters for efficient routing.

3 Methodology

3.1 Mapping Function from LID to Pruning Parameter

We start by briefly reviewing the notions of Local Intrinsic Dimensionality (LID) in the words of analysis. A full treatment can be found, for example, in Houle [17].

Definition 3.1 (Local Intrinsic Dimensionality)**.** Let $\mathcal{X}$ be a domain equipped with a distance measure $d : \mathcal{X} \times \mathcal{X} \to \mathbb{R}^+$. For a reference point $x \in \mathcal{X}$, let $F_x(r) = \mathbb{P}(d(x, Y) \leq r)$ denote the cumulative distribution function (CDF) of the distance between x and a random variable Y drawn from the underlying data distribution, where $\mathbb{P}$ denotes the probability function and r denotes the radius. The Local Intrinsic Dimensionality (LID) of x, denoted as $\mathrm{LID}(x)$, is defined as the intrinsic growth rate of the probability measure within the neighborhood of x:

$$\mathrm{LID}(x) \triangleq \lim_{r \to 0} \frac{r \cdot F'_x(r)}{F_x(r)} = \lim_{r \to 0} \frac{d \ln F_x(r)}{d \ln r}, \tag{1}$$

provided that the limit exists and $F_x(r)$ is continuously differentiable for $r > 0$.

While Definition 3.1 is elegant, it is often impractical to compute directly. According to [1], LID can be estimated as follows.

Definition 3.2 (LID Maximum Likelihood Estimator)**.** Given a reference point x and its k-nearest neighbors determined by the distance measure d. Let $r_i = d(x, v_i)$ denote the distance to the i-th nearest neighbor, sorted such that $r_1 \leq \cdots \leq r_k$. Following the formulation in [1], which adapts the Hill estimator for intrinsic dimensionality, the LID at x can be estimated as:

$$\widehat{\mathrm{LID}}(x) = -\left(\frac{1}{k} \sum_{i=1}^{k} \ln \frac{r_i}{r_k} \right)^{-1}. \tag{2}$$

The primary hypothesis of the proposed Manifold-Consistent Graph Indexing (MCGI) is that the graph topology should adapt to the local geometric complexity, rather than being a static vertex-edge structure. To preserve topological fidelity, the indexing algorithm must adopt a conservative pruning strategy (smaller α in [33]), thereby forcing the search to take smaller, safer steps along the manifold surface. The question then becomes how to quantitatively map the estimated LID to the pruning parameter α that controls the edge selection during graph construction. We refer to such a function as the *mapping function* $\Phi : \mathbb{R}^+ \to [\alpha_{\min}, \alpha_{\max}]$, which translates the geometric complexity of the local neighborhood into a corresponding pruning constraint.

Formally, let $u \in V$ be a node in the graph and $\widehat{\mathrm{LID}}(u)$ be its estimated LID. We define the *pruning parameter* $\alpha(u)$ as:

$$\alpha(u) \triangleq \Phi(\widehat{\mathrm{LID}}(u)), \tag{3}$$

where Φ will be constructed to satisfy the following geometric requirements: in regions with high LID, the graph should enforce a stricter connectivity constraint (smaller α) to avoid short-circuiting the manifold; conversely, in low-LID regions, the constraint can be relaxed (larger α).

To ensure the mapping is robust across datasets with varying complexity scales, we employ Z-score normalization based on the empirical distribution of the LID estimates. We first compute the normalized score $z(u)$:

$$z(u) = \frac{\widehat{\mathrm{LID}}(u) - \mu_{\widehat{\mathrm{LID}}}}{\sigma_{\widehat{\mathrm{LID}}}}, \tag{4}$$

where $\mu_{\widehat{\mathrm{LID}}}$ and $\sigma_{\widehat{\mathrm{LID}}}$ denote the mean and standard deviation of the set of estimated LID values $\{\widehat{\mathrm{LID}}(v) \mid v \in V\}$ computed across the entire graph. We then formulate Φ using a logistic function to smoothly map the Z-score to the operational range $[\alpha_{\min}, \alpha_{\max}]$:

$$\Phi(\widehat{\mathrm{LID}}(u)) = \alpha_{\min} + \frac{\alpha_{\max} - \alpha_{\min}}{1 + \exp(z(u))}. \tag{5}$$

The mapping function Φ satisfies the following geometric properties that are essential for stable graph construction: Monotonicity and Boundedness. We state these properties in the following as propositions with proof sketches and provide full proofs in the Appendix.

Proposition 3.3 (Monotonicity)**.** *The mapping function Φ is strictly decreasing with respect to the estimated local intrinsic dimensionality. Formally, given that the standard deviation of the LID estimates $\sigma_{\widehat{LID}} > 0$ and the pruning range $\alpha_{\max} > \alpha_{\min}$, the derivative satisfies:*

$$\frac{d\Phi}{d\widehat{LID}(u)} < 0. \tag{6}$$

Proof Sketch. By the chain rule, $\frac{d\Phi}{d\widehat{\text{LID}}(u)} = \frac{d\Phi}{dz} \cdot \frac{dz}{d\widehat{\text{LID}}(u)}$, where the Z-score satisfies $\frac{dz}{d\widehat{\text{LID}}(u)} = \frac{1}{\sigma_{\widehat{\text{LID}}}} > 0$. The derivative of the logistic component evaluates to $\frac{d\Phi}{dz} = -(\alpha_{\max} - \alpha_{\min}) \cdot \frac{e^z}{(1+e^z)^2}$, which is strictly negative since $\alpha_{\max} > \alpha_{\min}$ and $e^z > 0$ for all $z \in \mathbb{R}$. Combining these two terms yields a strictly negative derivative: $\alpha(u)$ decreases monotonically as geometric complexity increases. □

Proposition 3.4 (Boundedness)**.** *The pruning parameter $\alpha(u)$ derived from the mapping function is strictly bounded within the prescribed operational interval. That is, for any node u with a finite LID estimate, the following holds:*

$$\alpha_{\min} < \alpha(u) < \alpha_{\max}. \tag{7}$$

Proof Sketch. For any finite $\widehat{\text{LID}}(u)$, the Z-score $z(u)$ is finite, and the logistic function satisfies $S(u) = \frac{1}{1+e^{z(u)}} \in (0, 1)$ strictly, since $e^{z(u)} \in (0, \infty)$. Substituting into $\alpha(u) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot S(u)$ and applying the strict bounds on $S(u)$ directly yields $\alpha_{\min} < \alpha(u) < \alpha_{\max}$. □

3.2 Indexing Algorithms

The Manifold-Consistent Graph Indexing (MCGI) algorithm introduces a geometric calibration phase to a graph indexing procedure. Unlike static methods that apply a uniform connectivity rule, MCGI executes in two distinct stages to ensure that the topology respects the manifold structure, as follows. The full pseudo-code of the algorithm is in Appendix A.3, Algorithm 1.

Phase 1: Geometric Calibration. Before modifying the graph topology, the system first performs a global analysis of the dataset geometry. We estimate the LID for every point and aggregate the population statistics (μ, σ) defined in Section 3.1. This phase freezes the geometric profile of the dataset. By pre-computing these statistics, we decouple the complexity estimation from the graph update loop, ensuring that the mapping function Φ remains stable and computationally efficient during the intensive edge-selection process.

Phase 2: Manifold-Consistent Refinement. The index construction follows an iterative refinement strategy. Let $\mathcal{N}(u)$ denote the set of neighbors for node u in the graph G. In each iteration, the algorithm dynamically updates $\mathcal{N}(u)$ by considering a candidate pool $\mathcal{C}$ obtained from a greedy search on the current graph structure:

1. Queries the pre-computed geometric profile to determine the node-specific constraint $\alpha(u)$.
2. Explores the graph to identify a candidate pool $\mathcal{C}$.
3. Filters connections using the dynamic occlusion criterion.

Remark The calibration phase takes $O(N \log N)$ dominated by the construction phase, on par with the complexity of DiskANN [33]. In practice, calculating the exact LID for the entire dataset may cause a computational bottleneck on billion-scale data. To address this, we also design an online algorithm called *Online-MCGI*, whose pseudo-code can be found in Appendix A.3, Algorithm 2.

3.3 Theoretical Analysis

3.3.1 Geometric Complexity and Adaptive Routing

The efficiency of greedy routing on a proximity graph is intrinsically tied to the local topological properties of the underlying manifold $\mathcal{M}$. Under the Manifold Hypothesis, the search space is locally characterized by the Local Intrinsic Dimensionality (LID) at query q. We formalize the relationship between LID and the computational cost required to identify the nearest neighbor, as follows.

Lemma 3.5 (Local Complexity Lower Bound)**.** *Consistent with the complexity bounds established for growth-restricted metrics [23], for a query q on a manifold $\mathcal{M}$, the expected number of distance evaluations N_{dist} required for successful greedy routing scales exponentially with the local intrinsic dimensionality $d = LID(q)$:*

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right), \tag{8}$$

where $\lambda > 0$ is a geometric constant derived from the required convergence rate.

Proof Sketch. At a node u with distance r to query q, a greedy step succeeds only if a neighbor v falls within a cone of half-angle $\theta_{\max}$ around the direction of q; the probability of this event corresponds to the ratio of a spherical cap to the full hypersphere $\mathcal{S}^{d-1}$. Asymptotic analysis of this ratio via the regularized incomplete beta function yields $P_{\text{success}} \propto d^{-1/2}\exp(-\lambda \cdot d)$, where the exponential decay arises from the concentration of measure on high-dimensional spheres and $\lambda = -\ln(\sin\theta_{\max}) > 0$. Since $\mathbb{E}[N_{dist}] = 1/P_{\text{success}}$, the lower bound follows directly. □

Lemma 3.5 reveals a fundamental geometric barrier: for a fixed budget L, the probability of routing failure grows exponentially in high-LID regions. To address this, MCGI adopts an *Iso-Recall* strategy, aiming to homogenize the search reliability across the heterogeneous manifold.

Proposition 3.6 (Optimal Budget Allocation)**.** *To maintain a uniform bound on the routing failure probability δ across the manifold (i.e., $\mathbb{P}(\textit{fail} \mid q) \le \delta$ for all $q \in \mathcal{M}$), the search beam width $L(q)$ must scale exponentially with the local intrinsic dimensionality:*

$$L(q) \propto \exp\left(\lambda \cdot \textit{LID}(q)\right). \tag{9}$$

Proof Sketch. For a beam of width L, the failure probability is $(1 - P_{\text{success}})^L \le \delta$, which via the approximation $\ln(1-x) \approx -x$ requires $L \ge -\ln\delta / P_{\text{success}}$. Substituting the bound $P_{\text{success}} \propto d^{-1/2}\exp(-\lambda \cdot d)$ from Lemma 3.5 and absorbing the polynomial and confidence factors into a constant C_δ yields $L(q) \ge C_\delta \cdot \exp(\lambda \cdot \text{LID}(q))$. □

Proposition 3.6 implies that the search budget of each query should grow with the LID of its neighborhood. A change of L at every expansion step would disrupt memory alignment and the vectorized distance computation of the search loop; in contrast, a single resize of the candidate list per query leaves both intact. MCGI therefore realizes the budget of Proposition 3.6 at query time through the query-adaptive search list of Section 3.4, and it complements this budget at construction time through the pruning parameter $\alpha(u)$.

3.3.2 Topological Fidelity and Connectivity

A primary theoretical concern with aggressive edge pruning is the potential fracture of the connectivity backbone [39]. We will show that MCGI guarantees global reachability by strictly preserving the underlying manifold skeleton.

Formally, the edge selection in MCGI is governed by the adaptive pruning parameter $\alpha(u)$. Specifically, an edge (u, v) is pruned if there exists a witness node n such that $\alpha(u) \cdot d(n, v) \le d(u, v)$. This condition defines an *exclusion region* around the midpoint of u and v. As $\alpha(u)$ increases, this region shrinks, making pruning more conservative. We formulate this in the following proposition.

Proposition 3.7 (Connectivity Preservation)**.** *Let G_{EMST} and G_{RNG} denote the Euclidean Minimum Spanning Tree (EMST) and the Relative Neighborhood Graph (RNG), respectively. For any configuration of points in general position, provided that $\alpha(u) \ge 1.0$ for all $u \in V$, the graph G_{MCGI} satisfies the strict inclusion hierarchy:*

$$E_{EMST} \subseteq E_{RNG} \subseteq E_{MCGI}. \tag{10}$$

Consequently, G_{MCGI} is connected.

Proof Sketch. The first inclusion $E_{EMST} \subseteq E_{RNG}$ follows from the classical result of Toussaint [35], which holds for any finite point set in a metric space. For the second inclusion $E_{RNG} \subseteq E_{MCGI}$, note that MCGI prunes an edge (u, v) only if a witness node falls within a exclusion region of radius $\frac{1}{\alpha(u)}d(u, v)$ around v; since $\alpha(u) \ge 1.0$, this exclusion region is a strict geometric subset of the RNG exclusion region, so any edge surviving the RNG pruning condition is guaranteed to survive MCGI pruning. Connectivity of G_{MCGI} follows immediately from the inclusion hierarchy, since G_{EMST} spans all vertices by definition. □

Proposition 3.7 ensures that there are no structural dead ends. While Lemma 3.5 dictates the *cost* of routing, Proposition 3.7 ensures the *viability* of routing. Even in the worst-case scenario where heuristics fail, a path exists from the entry point to any target via the edges of the underlying EMST, provided the graph traversal algorithm has sufficient width to discover these backbone links.

3.3.3 Robustness to LID Estimation Error

A practical concern with the LID-based mapping is whether estimation noise from the finite-sample MLE (Definition 3.2) propagates into the pruning parameter $\alpha(u)$ in a destabilizing manner. We address this concern in the following.

The MLE estimator in Definition 3.2 is known to exhibit asymptotic variance of order LID^2/k [25], yielding a pointwise estimation error bounded as:

$$\left|\widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)\right| = O\left(\frac{\mathrm{LID}(u)}{\sqrt{k}}\right). \tag{11}$$

We characterize how this error propagates through the mapping function Φ.

Proposition 3.8 (Estimation Robustness)**.** *Let $\widehat{\mathrm{LID}}(u)$ be the MLE estimate computed with k nearest neighbors. The induced perturbation on the pruning parameter satisfies:*

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \mathrm{LID}(u)}{4\,\sigma_{\widehat{\mathrm{LID}}}\,\sqrt{k}}. \tag{12}$$

Proof Sketch. By a first-order Taylor expansion, $|\Delta\alpha(u)|$ is bounded by $\sup_L \left|\frac{d\Phi}{dL}\right| \cdot |\epsilon|$, where $\epsilon = \widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)$ is the estimation error. The gradient magnitude is bounded by $\frac{\alpha_{\max}-\alpha_{\min}}{4\,\sigma_{\widehat{\mathrm{LID}}}}$, since the logistic derivative $\frac{e^z}{(1+e^z)^2}$ achieves its global maximum of $\frac{1}{4}$ at $z = 0$, providing an intrinsic noise attenuation factor regardless of LID magnitude. Substituting the known asymptotic MLE error $|\epsilon| = O\left(\mathrm{LID}(u)/\sqrt{k}\right)$ [25] yields the bound directly, confirming that $|\Delta\alpha(u)|$ decays as $O(1/\sqrt{k})$ and even modest k suffices for stable α assignments. □

Proposition 3.8 reveals two structural properties. First, the perturbation decays as $O(1/\sqrt{k})$, confirming that even a modest neighborhood size (e.g., $k = 20$) provides stable α assignments. Second, the Sigmoid mapping contributes a factor of at most $1/4$ (the global maximum of the logistic derivative), acting as an intrinsic noise buffer regardless of the LID magnitude. This provides a principled lower bound on the required k during the geometric calibration phase.

3.3.4 Fixed-Beam Suboptimality and Adaptation

Proposition 3.6 establishes that an *ideal* search system allocates the beam width $L^*(q) \propto \exp(\lambda \cdot \mathrm{LID}(q))$ at query time. We now formalize the cost of a fixed beam width, which motivates both the query-adaptive search list of MCGI and its construction-time adaptation.

Definition 3.9 (Iso-Recall Beam Schedule)**.** For a target failure probability δ, the oracle beam schedule is defined as:

$$L^*(q) = C_\delta \cdot \exp(\lambda \cdot \mathrm{LID}(q)), \tag{13}$$

where $C_\delta = -\ln\delta / P_{\mathrm{success}}$ is a recall-dependent constant. A search algorithm is said to be *recall-optimal* if it implements $L^*(q)$ for every query q.

Proposition 3.10 (Suboptimality of Fixed Beam)**.** *For any fixed beam width $\bar{L}$, the recall gap in high-LID regions satisfies:*

$$P(\mathrm{fail} \mid q) - \delta \;\geq\; 1 - \exp\left(-\frac{C_\delta}{\bar{L}}\exp\big(\lambda\big(\mathrm{LID}(q) - \mathrm{LID}_{\mathrm{avg}}\big)\big)\right), \tag{14}$$

which grows exponentially with the deviation of $\mathrm{LID}(q)$ from the dataset mean $\mathrm{LID}_{\mathrm{avg}}$.

Proof Sketch. For a fixed beam width $\bar{L}$, the failure probability is $(1 - P_{\mathrm{success}})^{\bar{L}}$, while the oracle schedule $L^*(q) = C_\delta \exp(\lambda \cdot \mathrm{LID}(q))$ (Definition 3.9) achieves exactly δ by construction; the recall gap is thus the difference between these two quantities. In high-LID regions where $\mathrm{LID}(q) > \mathrm{LID}_{\mathrm{avg}}$, the shortfall $\Delta L(q) = L^*(q) - \bar{L} > 0$ grows exponentially with $\mathrm{LID}(q)$, and applying the inequality $1 - e^{-x} \leq x$ to the resulting expression yields the closed-form lower bound in Eq. (14). The right-hand side is monotonically increasing in $\mathrm{LID}(q)$, so the recall deficit grows exponentially, creating a widening gap that cannot be compensated by any fixed-parameter strategy. □

Proposition 3.10 quantifies precisely why static-parameter methods degrade on high-dimensional data: the recall deficit is not a constant offset but an *exponentially widening gap*. This provides a theoretical justification for geometry-aware adaptation that goes beyond empirical motivation. The query-adaptive search list of Section 3.4 reduces this gap directly by approximating $L^*(q)$ for each query. For a search that keeps a fixed beam width, the following corollary establishes that MCGI's construction-time adaptation monotonically reduces this gap compared to any fixed-parameter baseline, structurally compensating for the shortfall of a constant beam width.

Corollary 3.11 (Construction-time Structural Compensation for Fixed Beam Width)**.** *Under any fixed beam width $\bar{L}$, modulating $\alpha(u)$ during index construction according to $\Phi(\widehat{\mathrm{LID}}(u))$ monotonically reduces $P(\mathrm{fail} \mid q)$ for all queries in high-LID regions compared to any static-α baseline operating under the same $\bar{L}$. Specifically, nodes in high-LID regions receive conservative pruning (small α), preserving a strictly larger navigational edge set that structurally compensates for the shortfall of $\bar{L}$ relative to the oracle schedule $L^*(q)$. This guarantee is monotone but not tight: it establishes that MCGI dominates fixed-α baselines, rather than quantifying the residual gap to the Iso-Recall Schedule $L^*(q)$ (Definition 3.9) directly.*

Proof Sketch. For nodes where $\mathrm{LID}(u) > \mathrm{LID}_{\mathrm{avg}}$, Proposition 3.3 guarantees $\alpha(u) < \bar{\alpha}$, which strictly shrinks the pruning exclusion region and thus preserves a superset of edges relative to any static-$\bar{\alpha}$ baseline, i.e., $E_{\mathrm{MCGI}} \supseteq E_{\bar{\alpha}}$ in high-LID regions. A strictly larger outgoing edge set monotonically increases $P_{\mathrm{success}}(q)$ at each beam step by the same geometric argument as Proposition 3.6, yielding $P(\mathrm{fail} \mid q, \mathrm{MCGI}) \leq P(\mathrm{fail} \mid q, \bar{\alpha})$ for all q with $\mathrm{LID}(q) > \mathrm{LID}_{\mathrm{avg}}$. Note that this guarantee is monotone but not tight: quantifying the residual gap to the oracle schedule $L^*(q)$ (Definition 3.9) is left for future work. □

3.4 Query-Adaptive Search List

MCGI applies the iso-recall schedule of Definition 3.9 at query time, where it resizes the candidate list of each query according to the geometry that the search observes around the query. The schedule requires $\mathrm{LID}(q)$ before the search, although the neighborhood of q is unknown at that point. The beam search of a disk-resident Vamana index, however, maintains two structures that expose this neighborhood as the search proceeds. Specifically, the candidate list holds up to $\bar{L}$ nodes ordered by their compressed distances to q, and the expanded set records the exact distance of every node whose vector the search has read from disk. MCGI therefore inspects these two structures at a checkpoint after the search expands n_0 nodes, where n_0 is a small constant of a few tens, and resizes the candidate list once at this checkpoint.

At the checkpoint, MCGI forms a feature vector $\mathbf{f}(q)$ of five quantities that describe the local geometry around q. The first feature is $\ln \widehat{\mathrm{LID}}(q)$, which Definition 3.2 computes from the exact distances $r_1 \leq \cdots \leq r_k$ of the k nearest expanded nodes, where k equals the result size, typically 10. Because k distances yield a noisy estimate, the second feature applies the same estimator to the compressed distances of the m best candidates in the list, where m is a few tens. The next two features, $\ln r_1$ and $\ln r_k$, record the scale of the neighborhood, which the LID estimator removes by construction. The last feature is the logarithm of the number of candidates whose distance lies within $(1+\epsilon)\, r_k$ for a small constant ϵ such as 0.2. A large count indicates that many points compete for the last result positions, which is the concentration of distances that a high LID causes (Lemma 3.5).

MCGI maps the features to a list size through a log-linear score that mirrors the exponential form of Eq. (13):

$$L(q) = \min\Big\{ c\bar{L},\ \max\big\{ L_{\min},\ \bar{L} \exp\big(\gamma\, (\mathbf{w}^\top \mathbf{f}(q) + b - \bar{s})\big) \big\} \Big\}, \tag{15}$$

where $\bar{L}$ is the list size that the application requests, $\gamma > 0$ sets the strength of the adaptation, and c is a small cap such as 8. The weights $\mathbf{w}$ and the offset b result from a least-squares fit on a set of calibration queries that is disjoint from the evaluation queries; for each calibration query, the regression target is the logarithm of the smallest list size at which the query reaches its best recall. The constant $\bar{s}$ is the mean score of the calibration queries, so that a query of average geometry keeps the requested list size $\bar{L}$. A query whose features indicate a high LID consequently receives a longer list, whereas a query in a low-LID region terminates earlier and saves the disk reads that a fixed list would spend on it.

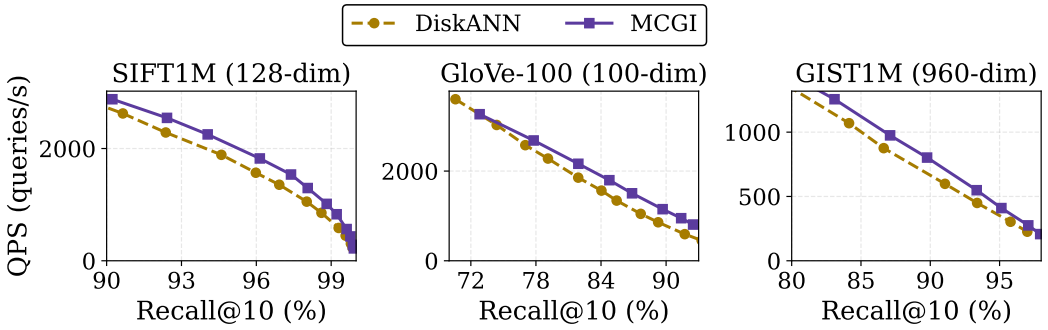


Figure 1: Recall-QPS trade-off of MCGI and DiskANN on the million-scale datasets.

The adaptation adds no disk reads, because every feature reuses distances that the search has already computed. Computing $\mathbf{f}(q)$ takes $O(n_0 + \bar{L})$ time once per query, which is small compared with the $O(\bar{L}R)$ compressed-distance evaluations of the search itself. Resizing the list either truncates its tail or raises its capacity, after which the beam search continues from its current state without a restart. The vectorized distance computation therefore remains unchanged, since the adaptation only moves the termination point of each query. Because the index format is also unchanged, the query-adaptive search list applies to any Vamana index, including an index built with the construction-time adaptation of Section 3.2.

4 Evaluation

The anonymized code is at: `https://github.com/hpdic/mcgi`.

All experiments run on a cloud virtual machine with 192 virtual cores of Intel Xeon Platinum 8468 and 960 GiB of RAM, where the disk-resident indexes reside on a 17.6 TB virtual block device (virtio). Each search reads graph nodes from this device with direct I/O, and at 32 search threads the measured I/O time is about 0.34 ms per read; consequently, the throughput of each method follows its number of disk reads per query. We evaluate MCGI on three million-scale benchmarks with dimensionality from 100 to 960, namely SIFT1M [21], GloVe-100 [29], and GIST1M [21], and on the billion-scale SIFT1B [22] and Deep1B [2]. The baseline is DiskANN (Vamana) [33], which searches the same index with a fixed candidate list. More setup details are in Appendix A.2.

4.1 Million-scale Effectiveness

We compare MCGI with DiskANN on a single index per dataset, so that the two methods differ only in the candidate list of each query. Each index is built with the standard Vamana pruning rule, and both methods search it with the same compressed vectors and the same requested list sizes $\bar{L}$ from 10 to 400. The weights of Eq. (15) come from 10,000 calibration queries per dataset, which are disjoint from the official evaluation queries. Each method runs once per list size on an otherwise idle machine, and the page cache is dropped before each run.

Figure 1 plots the query throughput against Recall@10 for the two methods. MCGI raises the throughput of DiskANN on all three million-scale datasets, and the improvement grows with the target recall. On SIFT1M, the gain rises from $1.15\times$ at 95% recall to $1.33\times$ at 99%. GloVe-100 and GIST1M follow the same trend, with gains from $1.12\times$ at 80% to $1.43\times$ at 90% and from $1.12\times$ at 85% to $1.20\times$ at 95%, respectively. The growth with recall agrees with Proposition 3.10, because a high target recall forces a fixed search to use a long list for every query, including the many queries in low-LID regions that converge early. MCGI shortens the lists of these queries and lengthens the lists of queries in high-LID regions, which removes disk reads that do not improve the result.

4.2 Billion-scale Scalability

We build one DiskANN index for each of SIFT1B and Deep1B with the standard sharded construction under a memory budget of 300 GiB. The construction splits the 8-bit vectors of SIFT1B into three

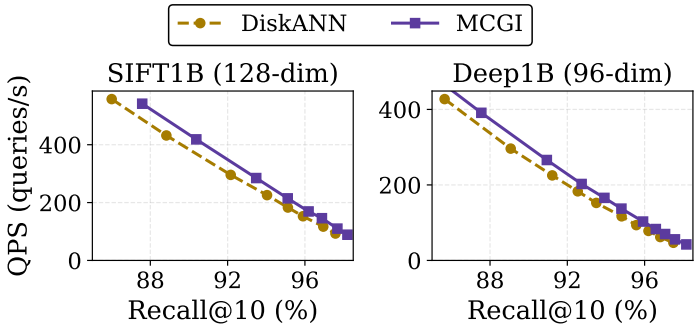


Figure 2: Recall-QPS trade-off of MCGI and DiskANN on the billion-scale datasets.

overlapping shards and completes in 4.8 hours on the CPU cores of the platform, while the 32-bit vectors of Deep1B need seven shards and 5.4 hours. Both indexes use $R = 32$, $L_{build} = 50$, and 31-byte PQ codes, and both methods again search the same index. The calibration queries of each dataset are the even-indexed half of its 10,000 official queries, and the evaluation uses the odd-indexed half. The requested list sizes extend to $\bar{L} = 1{,}000$ on SIFT1B and to $\bar{L} = 2{,}000$ on Deep1B, since DiskANN needs a list longer than 400 on SIFT1B and longer than 800 on Deep1B to reach 95% recall.

Figure 2 shows that the gain of MCGI persists at one billion vectors on both datasets and again grows with the target recall. On SIFT1B, MCGI raises the throughput of DiskANN from $1.07\times$ at 85% recall to $1.17\times$ at 95% and to $1.22\times$ at 97%, where the throughput rises from 115 to 141 queries per second. Deep1B follows the same trend with gains from $1.08\times$ at 85% to $1.18\times$ at 95% and to $1.24\times$ at 97%, where the throughput rises from 58 to 72 queries per second. At the longest list that we evaluate, DiskANN stays below 98% recall on both datasets, whereas MCGI reaches 98.2% on each of them, at 89 queries per second on SIFT1B and 42 on Deep1B.

5 Conclusion

This paper presents MCGI, a geometry-aware disk-resident indexing method for high-dimensional data of varying intrinsic dimensionality. MCGI addresses the Euclidean-Geodesic mismatch by adapting both the graph topology and the search budget to local intrinsic dimensionality. Specifically, MCGI estimates the LID around each query from the distances that its own search has already computed, and it resizes the candidate list of the query once, so that queries in high-LID regions receive longer lists and queries in low-LID regions terminate earlier. Our theoretical analysis shows that any fixed beam width leaves a recall gap that grows exponentially with LID, and the query-adaptive list approximates the iso-recall schedule that closes this gap without additional disk reads. On popular benchmarks of 96 to 960 dimensions and up to one billion vectors, MCGI raises the throughput of the state-of-the-art DiskANN by up to $1.43\times$ at matched high recall on the same index, and by up to $1.24\times$ on one billion vectors.

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A Technical appendices and supplementary material

A.1 Complete Proofs

Proposition A.1 (Monotonicity)**.** *The mapping function Φ is strictly decreasing with respect to the estimated local intrinsic dimensionality. Formally, given that the standard deviation of the LID estimates $\sigma_{\widehat{LID}} > 0$ and the pruning range $\alpha_{\max} > \alpha_{\min}$, the derivative satisfies:*

$$\frac{d\Phi}{d\widehat{LID}(u)} < 0. \tag{16}$$

Proof. Let $L = \widehat{\mathrm{LID}}(u)$ be the independent variable. We define the normalized Z-score z as a function of L:

$$z(L) = \frac{L - \mu_{\widehat{\mathrm{LID}}}}{\sigma_{\widehat{\mathrm{LID}}}}. \tag{17}$$

The mapping function is defined as:

$$\Phi(L) = \alpha_{\min} + \frac{C}{1 + \exp(z(L))}, \tag{18}$$

where $C = \alpha_{\max} - \alpha_{\min}$. Since we strictly require $\alpha_{\max} > \alpha_{\min}$, it follows that $C > 0$. To determine the sign of the gradient, we apply the chain rule:

$$\frac{d\Phi}{dL} = \frac{d\Phi}{dz} \cdot \frac{dz}{dL}. \tag{19}$$

First, we differentiate the Z-score term with respect to L:

$$\frac{dz}{dL} = \frac{1}{\sigma_{\widehat{\mathrm{LID}}}}. \tag{20}$$

Next, we differentiate the logistic component Φ with respect to z:

$$\frac{d\Phi}{dz} = \frac{d}{dz}\left(\alpha_{\min} + C(1 + e^z)^{-1}\right) \tag{21}$$

$$= C \cdot (-1) \cdot (1 + e^z)^{-2} \cdot \frac{d}{dz}(1 + e^z) \tag{22}$$

$$= -C \cdot \frac{e^z}{(1 + e^z)^2}. \tag{23}$$

Combining these terms yields the full derivative:

$$\frac{d\Phi}{dL} = -\frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \cdot \frac{e^z}{(1 + e^z)^2}. \tag{24}$$

We analyze the sign of each component:

- The operational range constant $C > 0$.
- The standard deviation $\sigma_{\widehat{\mathrm{LID}}} > 0$, assuming the dataset exhibits non-zero geometric variance.
- The exponential function $e^z > 0$ for all $z \in \mathbb{R}$.
- The denominator $(1 + e^z)^2 > 0$.

Therefore, the term $\frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \frac{e^z}{(1+e^z)^2}$ is strictly positive. The leading negative sign guarantees that $\frac{d\Phi}{dL} < 0$. This confirms that the pruning parameter α strictly decreases as the local geometric complexity increases, thereby enforcing a more conservative graph topology in high-LID regions to prevent topology distortion in complex regions. □

Proposition A.2 (Boundedness)**.** *The pruning parameter $\alpha(u)$ derived from the mapping function is strictly bounded within the prescribed operational interval. For any node u with a finite LID estimate:*

$$\alpha_{\min} < \alpha(u) < \alpha_{\max}. \tag{25}$$

Proof. Let $S(u)$ denote the logistic component of the mapping function:

$$S(u) = \frac{1}{1 + \exp(z(u))}. \tag{26}$$

For any finite input $\widehat{\mathrm{LID}}(u)$, the Z-score $z(u)$ is finite. The exponential function maps the real line to the positive real line, i.e., $\exp(z(u)) \in (0, \infty)$. Consequently, the denominator lies in the interval $(1, \infty)$. Taking the reciprocal yields the bounds for the logistic component:

$$0 < S(u) < 1. \tag{27}$$

Substituting $S(u)$ back into the definition of Φ:

$$\alpha(u) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot S(u). \tag{28}$$

Since $(\alpha_{\max} - \alpha_{\min}) > 0$, we can apply the inequality boundaries:

$$\alpha(u) > \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot 0 = \alpha_{\min}, \tag{29}$$

$$\alpha(u) < \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \cdot 1 = \alpha_{\max}. \tag{30}$$

This proves that the topology is strictly confined. The pruning behavior never exceeds the relaxation upper limit ($\alpha_{\max}$) and never becomes stricter than the lower limit ($\alpha_{\min}$), ensuring graph connectivity and preventing degree explosion. □

Lemma A.3 (Local Complexity Lower Bound)**.** *Consistent with the complexity bounds established for growth-restricted metrics [23], for a query q on a manifold $\mathcal{M}$, the expected number of distance evaluations N_{dist} required for successful greedy routing scales exponentially with the local intrinsic dimensionality $d = LID(q)$:*

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right), \tag{31}$$

where $\lambda > 0$ is a geometric constant derived from the required convergence rate.

Proof. Let the local dataset around a query q be modeled as a uniform distribution on a manifold of intrinsic dimension d. Consider a greedy routing step where the algorithm is currently at a node u with distance $r = \|u - q\|$.

1. Geometric Condition for Improvement. To make progress, the algorithm must identify a neighbor v such that the distance to the query is reduced by a factor ϵ (where $0 < \epsilon < 1$), i.e., $\|v - q\| \leq (1 - \epsilon)r$. Let $s = \|u - v\|$ be the distance between the current node and the candidate neighbor (the step size). Applying the Law of Cosines to triangle $\triangle uvq$, we have:

$$\|v - q\|^2 = r^2 + s^2 - 2rs\cos\theta, \tag{32}$$

where θ is the angle between the vector $\vec{uq}$ and $\vec{uv}$. The improvement condition $\|v-q\|^2 \leq (1-\epsilon)^2 r^2$ implies:

$$\begin{aligned} r^2 + s^2 - 2rs\cos\theta &\leq (1-\epsilon)^2 r^2, \\ 2rs\cos\theta &\geq r^2 + s^2 - r^2(1-\epsilon)^2, \\ \cos\theta &\geq \frac{r^2 + s^2 - r^2(1-\epsilon)^2}{2rs}. \end{aligned} \tag{33}$$

Assuming the step size is small relative to the distance (i.e., $s \ll r$), the right-hand side is dominated by $1 - (1-\epsilon)^2 \approx 2\epsilon$, yielding a critical angle threshold θ_{max} such that any successful neighbor must fall within the cone defined by $0 \leq \theta \leq \theta_{max}$.

2. Volume Concentration of the Improving Region. In the tangent space at u, the directions of neighbors are uniformly distributed on the unit hypersphere $\mathbb{S}^{d-1}$. The probability $P_{success}$ of finding a neighbor in the improving cone corresponds to the ratio of the surface area of the spherical cap $\mathcal{C}(\theta_{max})$ to the total surface area of $\mathbb{S}^{d-1}$. This ratio is given by the regularized incomplete beta function, which can be expressed in terms of the integral over the polar angle ϕ:

$$P_{success} = \frac{\text{Area}(\mathcal{C}(\theta_{max}))}{\text{Area}(\mathbb{S}^{d-1})} = \frac{\int_0^{\theta_{max}} \sin^{d-2}\phi \, d\phi}{\int_0^{\pi} \sin^{d-2}\phi \, d\phi}. \tag{34}$$

3. Asymptotic Analysis. We analyze the asymptotic behavior of this ratio as $d \to \infty$. The denominator relates to the surface area of the hypersphere and can be evaluated using the standard integral identity for powers of sine. Exploiting the symmetry of $\sin\phi$ around $\pi/2$, we have:

$$\begin{aligned}\int_0^{\pi} \sin^{d-2}\phi\, d\phi &= 2\int_0^{\frac{\pi}{2}} \sin^{d-2}\phi\, d\phi \\ &= 2\cdot\frac{\sqrt{\pi}}{2}\frac{\Gamma(\frac{d-1}{2})}{\Gamma(\frac{d}{2})} = \sqrt{\pi}\frac{\Gamma(\frac{d-1}{2})}{\Gamma(\frac{d}{2})}.\end{aligned} \tag{35}$$

For high-dimensional spaces ($d \gg 1$), we apply the asymptotic property of the Gamma function ratio, $\frac{\Gamma(x+a)}{\Gamma(x)} \approx x^a$. Setting $x = d/2$ and $a = -1/2$, we obtain the approximation:

$$\sqrt{\pi}\frac{\Gamma(\frac{d}{2}-\frac{1}{2})}{\Gamma(\frac{d}{2})} \approx \sqrt{\pi}\left(\frac{d}{2}\right)^{-\frac{1}{2}} = \sqrt{\frac{2\pi}{d}}. \tag{36}$$

For the numerator, we exploit the concentration of measure near the integration boundary θ_{max}. Since the integrand $\sin^{d-2}\phi$ decays exponentially fast away from θ_{max}, the integral is dominated by the contribution near the upper limit. We apply integration by parts to extract the leading order term. Recall that $\frac{d}{d\phi}(\sin^{d-1}\phi) = (d-1)\sin^{d-2}\phi\cos\phi$. We rewrite the integrand as:

$$\int_0^{\theta_{max}} \sin^{d-2}\phi\, d\phi = \int_0^{\theta_{max}} \frac{1}{(d-1)\cos\phi}\frac{d}{d\phi}(\sin^{d-1}\phi)\, d\phi. \tag{37}$$

Neglecting lower-order terms generated by the derivative of $\frac{1}{\cos\phi}$, the integral approximates to the boundary term:

$$\left[\frac{\sin^{d-1}\phi}{(d-1)\cos\phi}\right]_0^{\theta_{max}} = \frac{\sin^{d-1}\theta_{max}}{(d-1)\cos\theta_{max}}. \tag{38}$$

Substituting these approximations back into the probability ratio, we obtain:

$$P_{success} \approx \frac{1}{\sqrt{2\pi}\cos\theta_{max}}\cdot\frac{\sqrt{d}}{d-1}\cdot(\sin\theta_{max})^{d-1}. \tag{39}$$

For large d, the term $\frac{\sqrt{d}}{d-1}$ is asymptotically equivalent to $d^{-1/2}$. Since θ_{max} is fixed by the geometric constraints, the trigonometric factors are constant relative to d. Let $\gamma = \sin\theta_{max} < 1$. The probability scales as:

$$P_{success} \propto d^{-1/2}\cdot\gamma^d = d^{-1/2}\exp(d\ln\gamma). \tag{40}$$

Defining the decay rate $\lambda = -\ln\gamma > 0$ (since $\gamma < 1$), we strictly obtain:

$$P_{success} \propto d^{-1/2}\exp(-\lambda\cdot d). \tag{41}$$

4. Complexity Bound. The expected number of distance evaluations follows $\mathbb{E}[N_{dist}] = 1/P_{success}$. Thus:

$$\mathbb{E}[N_{dist}] \geq \Omega\left(\sqrt{d}\cdot\exp(\lambda\cdot d)\right). \tag{42}$$

This confirms that the routing cost is dominated by the exponential term $\exp(\lambda\cdot\mathrm{LID}(q))$, necessitating the adaptive beam width design in MCGI. □

Proposition A.4 (Optimal Budget Allocation)**.** *To maintain a uniform bound on the routing failure probability δ across the manifold (i.e., $\mathbb{P}(\text{fail} \mid q) \leq \delta$ for all $q \in \mathcal{M}$), the search beam width $L(q)$ must scale exponentially with the local intrinsic dimensionality:*

$$L(q) \propto \exp\left(\lambda\cdot \textit{LID}(q)\right). \tag{43}$$

Proof. Consider a beam search with width L. The probability of failing to find a descent step in one round is approximately $(1 - P_{success})^L$. To guarantee a recall target, we enforce a constant upper bound δ on the failure probability:

$$(1 - P_{success})^L \leq \delta. \tag{44}$$

Taking the natural logarithm and using the approximation $\ln(1-x) \approx -x$ for small $P_{success}$, we require:

$$-L \cdot P_{success} \leq \ln \delta \implies L \geq \frac{-\ln \delta}{P_{success}}. \tag{45}$$

Substituting the relationship $P_{success} = 1/\mathbb{E}[N_{dist}]$ established in Lemma 3.5, we obtain the asymptotic upper bound $P_{success} \leq \mathcal{O}(\sqrt{d} \cdot \exp(-\lambda \cdot d))$, where $d = \text{LID}(q)$. To satisfy the recall constraint in the worst case scenario, the search budget L must grow inversely to $P_{success}$:

$$L \geq \frac{-\ln \delta}{\mathcal{O}(\sqrt{d} \cdot \exp(-\lambda \cdot d))} = \Omega\left(\frac{1}{\sqrt{d}} \cdot \exp(\lambda \cdot d)\right). \tag{46}$$

In high-dimensional spaces ($d \gg 1$), the exponential growth term strictly dominates the polynomial decay factor $1/\sqrt{d}$. By absorbing the lower-order polynomial variations and the confidence factor $-\ln \delta$ into a generalized constant C_δ, the necessary budget asymptotically simplifies to:

$$L(q) \geq C_\delta \cdot \exp(\lambda \cdot \text{LID}(q)). \tag{47}$$

□

Proposition A.5 (Connectivity Preservation)**.** *Let G_{EMST} and G_{RNG} denote the Euclidean Minimum Spanning Tree and the Relative Neighborhood Graph, respectively. For any configuration of points in general position, provided that $\alpha(u) \geq 1.0$ for all $u \in V$, the graph G_{MCGI} satisfies the strict inclusion hierarchy:*

$$E_{EMST} \subseteq E_{RNG} \subseteq E_{MCGI}. \tag{48}$$

Consequently, G_{MCGI} is connected.

Proof. The proof proceeds in two steps. First, we invoke the classical result established by Toussaint [35], which proves that the Relative Neighborhood Graph (RNG) is a supergraph of the Euclidean Minimum Spanning Tree (EMST) for any finite set of points in a metric space:

$$E_{EMST} \subseteq E_{RNG}. \tag{49}$$

Since the EMST connects all vertices in a single component, G_{RNG} is necessarily connected.

Second, we show that $E_{RNG} \subseteq E_{MCGI}$. In the RNG, an edge between u and v is pruned if and only if a witness node falls within the intersection of two spheres centered at u and v with radius $d(u, v)$. We denote this geometric intersection as the exclusion region $\mathcal{R}_{RNG}$. Therefore, an edge exists in the RNG if and only if $\mathcal{R}_{RNG}$ contains no data points. In MCGI, the pruning condition requires the witness node to be within a distance of $\frac{1}{\alpha(u)} d(u, v)$ from v. Because the algorithm enforces $\alpha(u) \geq 1.0$, the pruning radius in MCGI is strictly less than or equal to the pruning radius of the RNG. Consequently, the exclusion region of MCGI is a geometric subset of the RNG exclusion region:

$$\mathcal{R}_{MCGI} \subseteq \mathcal{R}_{RNG}. \tag{50}$$

This subset relationship of exclusion regions creates an inverse superset relationship for the preserved edges. If an edge survives the RNG condition, its $\mathcal{R}_{RNG}$ is empty. Since $\mathcal{R}_{MCGI}$ is contained entirely within $\mathcal{R}_{RNG}$, it must also be empty. Thus, any edge present in the RNG is guaranteed to be preserved in MCGI, yielding the inclusion hierarchy of edge sets:

$$E_{RNG} \subseteq E_{MCGI}. \tag{51}$$

By combining this with the classical result that $E_{EMST} \subseteq E_{RNG}$ derived in [35], we establish the complete global connectivity chain. □

Proposition A.6 (Estimation Robustness)**.** *Let $\widehat{\text{LID}}(u)$ be the MLE estimate computed with k nearest neighbors. The induced perturbation on the pruning parameter satisfies:*

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \text{LID}(u)}{4\, \sigma_{\widehat{\text{LID}}} \sqrt{k}}. \tag{52}$$

Proof. Let $\epsilon = \widehat{\mathrm{LID}}(u) - \mathrm{LID}(u)$ denote the estimation error. By the first-order Taylor expansion of Φ around $\mathrm{LID}(u)$:

$$|\Delta\alpha(u)| = \left|\Phi\Big(\widehat{\mathrm{LID}}(u)\Big) - \Phi(\mathrm{LID}(u))\right| \leq \sup_L \left|\frac{d\Phi}{dL}\right| \cdot |\epsilon|. \tag{53}$$

From the derivative computed in Proof A.1, the magnitude of the gradient is:

$$\left|\frac{d\Phi}{dL}\right| = \frac{C}{\sigma_{\widehat{\mathrm{LID}}}} \cdot \frac{e^z}{(1+e^z)^2}, \tag{54}$$

where $C = \alpha_{\max} - \alpha_{\min}$. The logistic factor $\frac{e^z}{(1+e^z)^2}$ achieves its global maximum of $1/4$ at $z = 0$, so:

$$\left|\frac{d\Phi}{dL}\right| \leq \frac{\alpha_{\max} - \alpha_{\min}}{4\,\sigma_{\widehat{\mathrm{LID}}}}. \tag{55}$$

Substituting the asymptotic error bound from Eq. (11), namely $|\epsilon| = O(\mathrm{LID}(u)/\sqrt{k})$ [25], yields:

$$|\Delta\alpha(u)| \leq \frac{(\alpha_{\max} - \alpha_{\min}) \cdot \mathrm{LID}(u)}{4\,\sigma_{\widehat{\mathrm{LID}}}\,\sqrt{k}}. \tag{56}$$

This completes the proof. □

Proposition A.7 (Suboptimality of Fixed Beam)**.** *For any fixed beam width $\bar{L}$, the recall gap in high-LID regions satisfies Eq.* (14)*, which grows exponentially with the deviation of* $\mathrm{LID}(q)$ *from the dataset mean.*

Proof. Consider a beam search with fixed width $\bar{L}$. From Proof A.4, the failure probability at query q is:

$$P(\text{fail} \mid q) = \left(1 - P_{\text{success}}(q)\right)^{\bar{L}}. \tag{57}$$

From Lemma 3.5, we have:

$$P_{\text{success}}(q) \propto d^{-1/2} \exp(-\lambda \cdot d), \quad d = \mathrm{LID}(q). \tag{58}$$

For the oracle schedule $L^*(q)$ in Definition 3.9, the constraint $P(\text{fail} \mid q) \leq \delta$ holds by construction. For fixed $\bar{L}$, define $\Delta L(q) = L^*(q) - \bar{L}$. In high-LID regions where $\mathrm{LID}(q) > \mathrm{LID}_{\text{avg}}$, we have $\Delta L(q) > 0$ and:

$$\begin{aligned} P(\text{fail} \mid q) - \delta &= (1 - P_{\text{success}})^{\bar{L}} - (1 - P_{\text{success}})^{L^*(q)} \\ &\geq 1 - \exp\left(-\frac{C_\delta}{\bar{L}} \exp\big(\lambda\big(\mathrm{LID}(q) - \mathrm{LID}_{\text{avg}}\big)\big)\right), \end{aligned} \tag{59}$$

where the inequality applies $1 - e^{-x} \leq x$ and substitutes $L^*(q)$ from Definition 3.9. The right-hand side is monotonically increasing in $\mathrm{LID}(q)$, confirming the exponential widening of the recall gap. □

Corollary A.8 (Construction-time Structural Compensation)**.** *Under any fixed beam width $\bar{L}$, the adaptive pruning in MCGI monotonically reduces $P(\text{fail} \mid q)$ for queries in high-LID regions compared to any static-α baseline operating under the same $\bar{L}$.*

Proof. We proceed in two steps: first showing that MCGI preserves a strictly larger edge set in high-LID regions, then connecting this to a lower failure probability under fixed beam search.

Step 1: Edge set inclusion. The pruning criterion in MCGI discards an edge (u, v) if and only if there exists a witness node n such that $\alpha(u) \cdot d(n, v) \leq d(u, v)$. For nodes where $\mathrm{LID}(u) > \mathrm{LID}_{\text{avg}}$, Proposition A.1 guarantees $\alpha(u) < \bar{\alpha}$. Since $\alpha(u) < \bar{\alpha}$ strictly shrinks the left-hand side of the pruning condition, any edge that would be pruned under $\alpha(u)$ is also pruned under $\bar{\alpha}$, but not vice versa. Therefore $\mathcal{E}_{\text{MCGI}} \supseteq \mathcal{E}_{\bar{\alpha}}$ in high-LID regions, with strict inclusion whenever at least one edge survives $\alpha(u)$ but not $\bar{\alpha}$.

Step 2: Failure probability. Under beam search with fixed width $\bar{L}$, the failure probability at query q is governed by $P_{\text{success}}(q)$, the probability of finding a descent step from any node on the routing path (Proof A.4). A strictly larger outgoing edge set at each high-LID node monotonically increases

the number of candidate directions evaluated per beam step, which by the same geometric argument in Proof A.4 yields a weakly higher $P_{\text{success}}(q)$ and thus:

$$P(\text{fail} \mid q, \text{MCGI}) \leq P(\text{fail} \mid q, \bar{\alpha}), \quad \forall\, q \text{ s.t. } \text{LID}(q) > \text{LID}_{\text{avg}}, \tag{60}$$

with equality only when $\alpha(u) = \bar{\alpha}$ for all u on the routing path, i.e., when the data is geometrically uniform. This establishes a monotone improvement over fixed-α baselines; quantifying the residual gap to the Iso-Recall Schedule $L^*(q)$ (Definition 3.9) is left for future work. □

A.2 Additional Experimental Details

Experiment Platform All experiments run on a cloud virtual machine with 192 virtual cores of Intel Xeon Platinum 8468 and 960 GiB of RAM. The operating system is Ubuntu 24.04 LTS, and all code is compiled with GCC 13.3 at optimization level -O3. The disk-resident indexes reside on a 17.6 TB virtual block device (virtio) of the virtual machine, which the DiskANN search engine reads through Linux asynchronous I/O with direct I/O, so that the operating system page cache does not serve graph nodes. Every search uses 32 threads and a beam width of 2, and the page cache is dropped before each search run. All indexes are built on the CPU cores with the DiskANN builder; the SIFT1B and Deep1B builds use a memory budget of 300 GiB and 192 threads.

Datasets We evaluate our method on three standard million-scale benchmarks whose dimensionality ranges from 100 to 960 and on two billion-scale benchmarks, so that the datasets cover both high ambient dimensionality and large scale. In the following descriptions, N denotes the number of base vectors and D represents the vector dimensionality:

- SIFT1M [21] ($N = 10^6, D = 128$): Local image descriptors with 10,000 queries, searched with Euclidean distance (L_2).
- GloVe-100 [29] ($N = 1.18 \times 10^6, D = 100$): Word embedding vectors with 10,000 queries. Following standard practice, we normalize the vectors to unit length and use Euclidean distance as a proxy for cosine similarity.
- GIST1M [21] ($N = 10^6, D = 960$): Global image descriptors with 1,000 queries, which have the highest ambient dimensionality and the widest spread of LID values among the million-scale datasets.
- SIFT1B [22] ($N = 10^9, D = 128$): One billion local image descriptors stored as 8-bit unsigned integers, with 10,000 queries and the official ground truth over the full billion vectors.
- Deep1B [2] ($N = 10^9, D = 96$): One billion image descriptors from a convolutional neural network, reduced to 96 dimensions by PCA, normalized to unit length, and stored as 32-bit floats, with 10,000 queries. The official ground truth over the full billion vectors is wrong for three queries (indices 449, 2382, and 2809)[1], which we exclude.

The calibration queries of the query-adaptive search list are disjoint from the evaluation queries. For SIFT1M and GIST1M, they are the first 10,000 vectors of the learning set of each dataset. GloVe-100 has no learning set; its calibration queries are 10,000 base vectors drawn with a fixed seed, and each of them is excluded from its own ground truth. The ground truth of every calibration query consists of its exact 100 nearest neighbors. For SIFT1B and Deep1B, the calibration queries are the even-indexed half of the official queries, and the evaluation uses the odd-indexed half, which leaves 5,000 evaluation queries on SIFT1B and 4,998 on Deep1B.

Baselines We compare MCGI against DiskANN (Vamana) [33], the standard disk-resident graph index. Both methods search the same index file with the same compressed vectors, the same beam width, and the same requested list sizes. The comparison therefore isolates the query-adaptive search list of MCGI, since the index, the I/O path, and the distance computation are identical.

Evaluation Metrics We adhere to the standard evaluation protocol for ANN search and measure Recall@10 against QPS (queries per second). Each method yields one Recall@10 and QPS point per requested list size. The throughput gain at a target recall is the ratio of the QPS of the two methods, where each QPS value is linearly interpolated along the recall axis of its curve.

[1] `https://github.com/harsha-simhadri/big-ann-benchmarks/issues/353`

Table 1: Dataset specifications and indexing parameters.

Dataset	N	D	Queries	R	L_{build}	m_{PQ} (Byte)	Data Type
SIFT1M	10^6	128	10,000	32	50	107	float32
GloVe-100	1.18×10^6	100	10,000	64	100	90	float32
GIST1M	10^6	960	1,000	32	100	322	float32
SIFT1B	10^9	128	5,000	32	50	31	uint8
Deep1B	10^9	96	4,998	32	50	31	float32

Table 2: LID statistics and search parameters.

Dataset	μ_{LID} (Mean)	σ_{LID} (Std)	List size $\bar{L}$	γ	Threads
SIFT1M	16.1	9.2	$10 \rightarrow 400$	0.7	32
GloVe-100	35.3	18.1	$10 \rightarrow 400$	1.0	32
GIST1M	48.9	25.9	$10 \rightarrow 400$	0.7	32
SIFT1B	–	–	$10 \rightarrow 1000$	0.7	32
Deep1B	–	–	$10 \rightarrow 2000$	0.7	32

Build Configuration Table 1 summarizes the dataset specifications and the indexing parameters of the index that both methods search, where the query count is the number of evaluation queries. R denotes the maximum outdegree of each node in the graph, and L_{build} is the size of the candidate list during index construction. Every index uses the standard pruning parameter $\alpha = 1.2$ of DiskANN. m_{PQ} specifies the number of bytes of the Product Quantization (PQ) code of each vector, which follows from the PQ memory budget of the build.

Search Configuration Table 2 lists the LID statistics of each dataset and the parameters of the query-adaptive search list. μ_{LID} and σ_{LID} denote the mean and standard deviation of the maximum-likelihood LID estimates of the base vectors with 20 neighbors, which we compute for the million-scale datasets only. The checkpoint of Eq. (15) follows $n_0 = 24$ expansions, the candidate-list estimator uses the $m = 64$ best candidates, and the competition count uses $\epsilon = 0.2$. The list size is bounded by $L_{\min} = 10$ and $c = 8$, and the strength γ is chosen from $\{0.5, 0.7, 1.0, 1.3\}$ on the calibration queries of each dataset.

Offline Cost Table 3 lists the offline cost of each index, where every build runs alone on the platform with 192 threads and the calibration runs alone with 32 search threads. The training of the PQ codebooks dominates the build of the million-scale indexes, whereas the graph construction over the shards and their merge dominates at one billion vectors. The only offline step that MCGI adds is the calibration of Eq. (15), which searches the calibration queries once at each list size of a fixed grid and then fits the weights by ridge regression within seconds. The grid has 14 list sizes from 10 to 400 for the million-scale datasets and SIFT1B, and 18 list sizes from 10 to 1,000 for Deep1B. Calibration takes about 3 minutes on each million-scale dataset, and 1.6 and 4.8 minutes on SIFT1B and Deep1B, which is at most 1.5% of the build time of a billion-scale index.

A.3 Algorithm Pseudo-Code

Algorithm 1 outlines the offline construction process of the indexing structure. The procedure is divided into two sequential phases.

- The first phase focuses on geometric calibration by calculating the Local Intrinsic Dimensionality for all vectors in the dataset in parallel. After obtaining the global mean and standard deviation of these estimates, the algorithm assigns a personalized pruning parameter α_u to each node. This parameter is derived using a logistic function that maps the standardized geometric complexity of the local neighborhood into a bounded operational range.
- The second phase handles topology refinement. Starting from a random graph, the algorithm iteratively optimizes the connections. For each node, it executes a greedy search to identify a pool of candidate neighbors. It then evaluates these candidates alongside existing neighbors using a dynamic occlusion criterion. A candidate is pruned if a previously accepted neighbor

Table 3: Offline cost in seconds: index build stages and calibration of the query-adaptive search list.

Dataset	PQ training	Graph	Disk layout	Total build	Calibration
SIFT1M	195	6	1	203	174
GloVe-100	400	15	1	416	168
GIST1M	1,678	82	4	1,764	174
SIFT1B	2,775	14,117	290	17,225	94
Deep1B	3,373	15,029	836	19,317	289

falls within a restricted exclusion region defined by the precomputed α_u. Survivors are added to the updated neighbor list until the degree limit is reached.

Algorithm 1 Manifold-Consistent Graph Indexing (MCGI)

Input: Dataset X, Max Degree R, Beam Width L
Output: Optimized Graph G

// Phase 1: Geometric Calibration
$\mathcal{L} \leftarrow$ ParallelEstimateLID(X)
$\mu \leftarrow$ Mean($\mathcal{L}$)
$\sigma \leftarrow$ StdDev($\mathcal{L}$)
for each node $u \in V$ in parallel **do**
  $z_u \leftarrow (\mathcal{L}[u] - \mu)/\sigma$
  $\alpha_u \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})/(1 + \exp(z_u))$
end for

// Phase 2: Topology Refinement
$G \leftarrow$ RandomGraph(X, R)
for $iter \leftarrow 1$ to $MaxIter$ **do**
  for each node $u \in G$ in parallel **do**
    $\mathcal{C} \leftarrow$ GreedySearch(u, G, L)
    $\mathcal{N}_{new} \leftarrow \emptyset$
    for $v \in$ SortByDistance($\mathcal{C} \cup \mathcal{N}(u)$) **do**
      $pruned \leftarrow$ False
      for $n \in \mathcal{N}_{new}$ **do**
        if $\alpha_u \cdot d(n, v) \leq d(u, v)$ **then**
          $pruned \leftarrow$ True; break
        end if
      end for
      if not $pruned$ **and** $|\mathcal{N}_{new}| < R$ **then**
        $\mathcal{N}_{new}$.add(v)
      end if
    end for
    $\mathcal{N}(u) \leftarrow \mathcal{N}_{new}$
  end for
end for

Algorithm 2 presents an optimized online variant designed to handle massive datasets where exhaustive geometric profiling is computationally prohibitive.

- It begins with a fast approximation phase that samples a small subset of the data. By evaluating this subset, the algorithm quickly bootstraps the global mean and standard deviation of the underlying probability distribution. This step provides a coarse but efficient estimate of the overall geometric complexity, enabling the algorithm to initialize the pruning parameters without incurring the cost of a full dataset pass.
- The second phase merges the geometric estimation directly into the iterative topology refinement loop. Instead of computing all complexities beforehand, the algorithm dynamically estimates the node specific dimensionality on the fly using the candidate set discovered

during the greedy search step. This real time estimate is immediately used to calculate the specific pruning parameter α_u. The algorithm then applies this dynamic parameter to the pruning criterion, determining which edges to preserve and which to discard. This approach ensures the graph connectivity adapts to the manifold geometry progressively as the quality of the candidate neighbors improves with each iteration.

Algorithm 2 Online Manifold-Consistent Graph Indexing

Input: Dataset X, Max Degree R, Beam Width L, Sample Rate S
Output: Optimized Graph G

// Phase 1: Bootstrap Statistics (Fast Approximation)
$X_{sample} \leftarrow$ RandomSample(X, S)
$\mathcal{L}_{sample} \leftarrow$ EstimateLID(X_{sample}) {Compute LID on subset}
$\mu \leftarrow$ Mean$(\mathcal{L}_{sample})$, $\sigma \leftarrow$ StdDev$(\mathcal{L}_{sample})$

// Phase 2: Online Topology Refinement
$G \leftarrow$ RandomGraph(X, R)
for $iter \leftarrow 1$ to $MaxIter$ **do**
  for each node $u \in G$ in parallel **do**
    $\mathcal{C} \leftarrow$ GreedySearch(u, G, L)
    // Online LID Estimation & Parameter Adaptation
    $\hat{\ell}_u \leftarrow$ ComputeLID$(u, \mathcal{C})$ {Estimate from neighbors}
    $z_u \leftarrow (\hat{\ell}_u - \mu)/\sigma$
    $\alpha_u \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})/(1 + \exp(z_u))$
    $\mathcal{N}_{new} \leftarrow \emptyset$
    for $v \in$ SortByDistance$(\mathcal{C} \cup \mathcal{N}(u))$ **do**
      $pruned \leftarrow$ False
      for $n \in \mathcal{N}_{new}$ **do**
        // Pruning with dynamic α_u
        if $\alpha_u \cdot d(n, v) \leq d(u, v)$ **then**
          $pruned \leftarrow$ True; break
        end if
      end for
      if not $pruned$ **and** $|\mathcal{N}_{new}| < R$ **then**
        $\mathcal{N}_{new}$.add(v)
      end if
    end for
    $\mathcal{N}(u) \leftarrow \mathcal{N}_{new}$
  end for
end for